

Modul ke

02

Fakultas

Teknik

Program Studi

Informatika

STEP 01 = KNOW USER

- REVIEW UCD,
- PERSONA USER

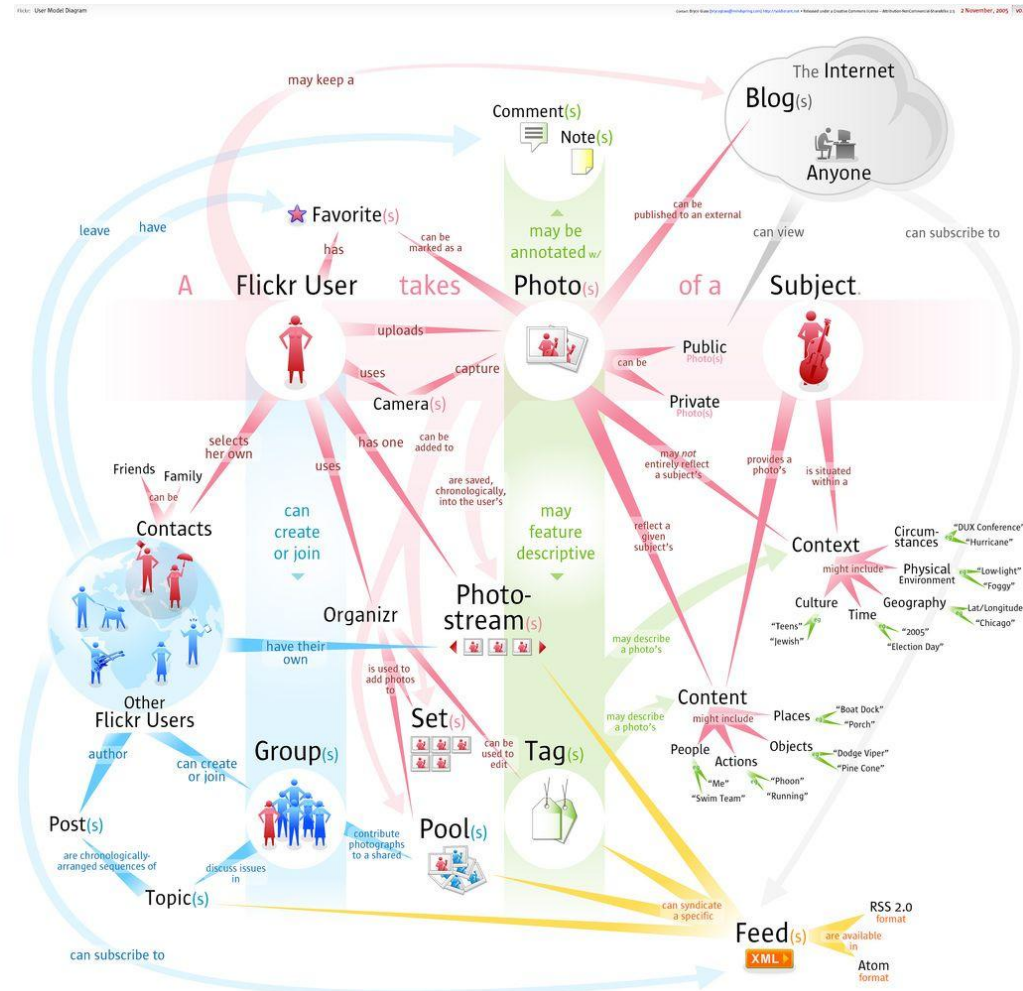
3 sks

Oleh :
Tim Dosen IMK

User Experience

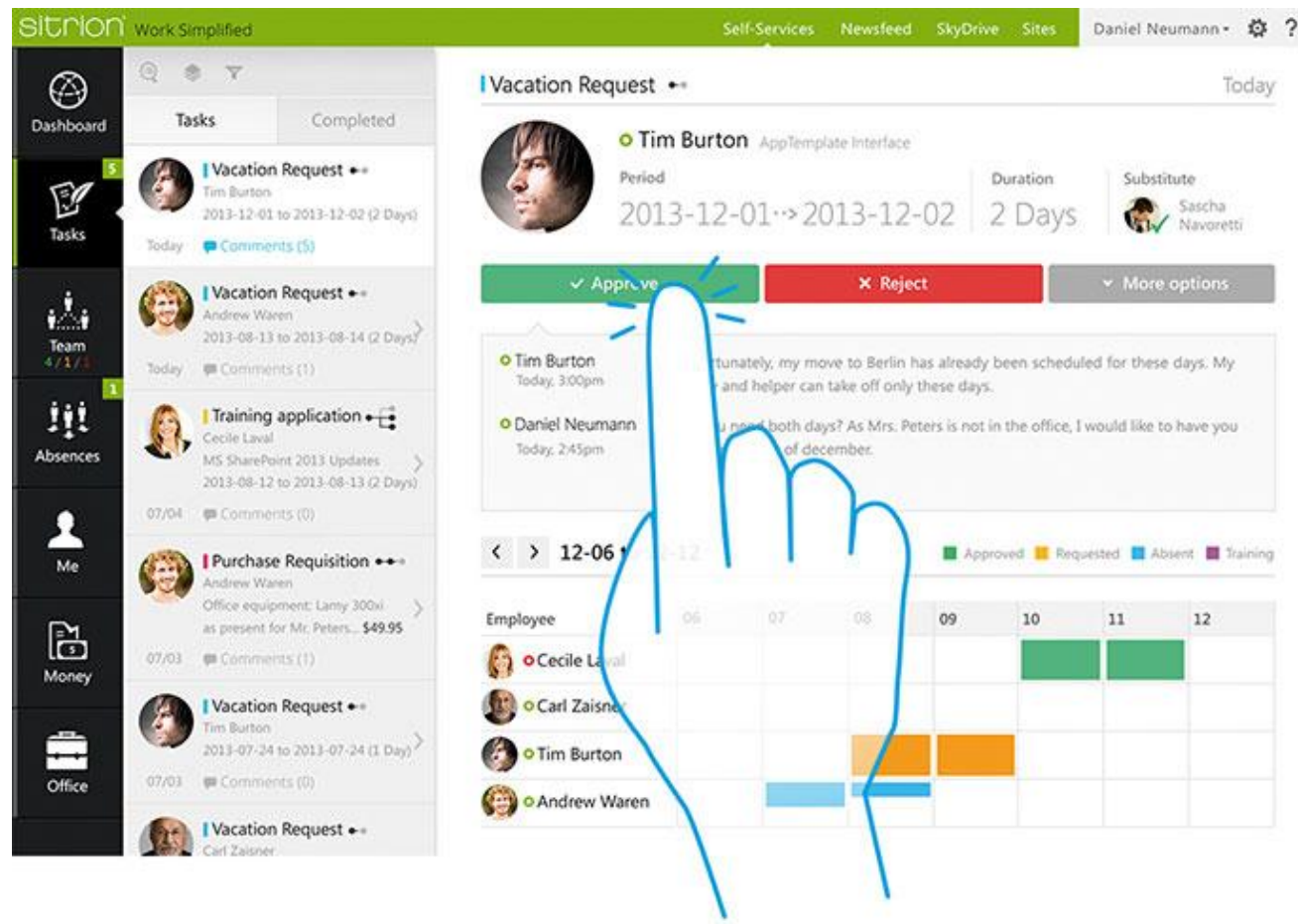


<https://uxmag.com/articles/cubi-a-user-experience-model-for-project-success>



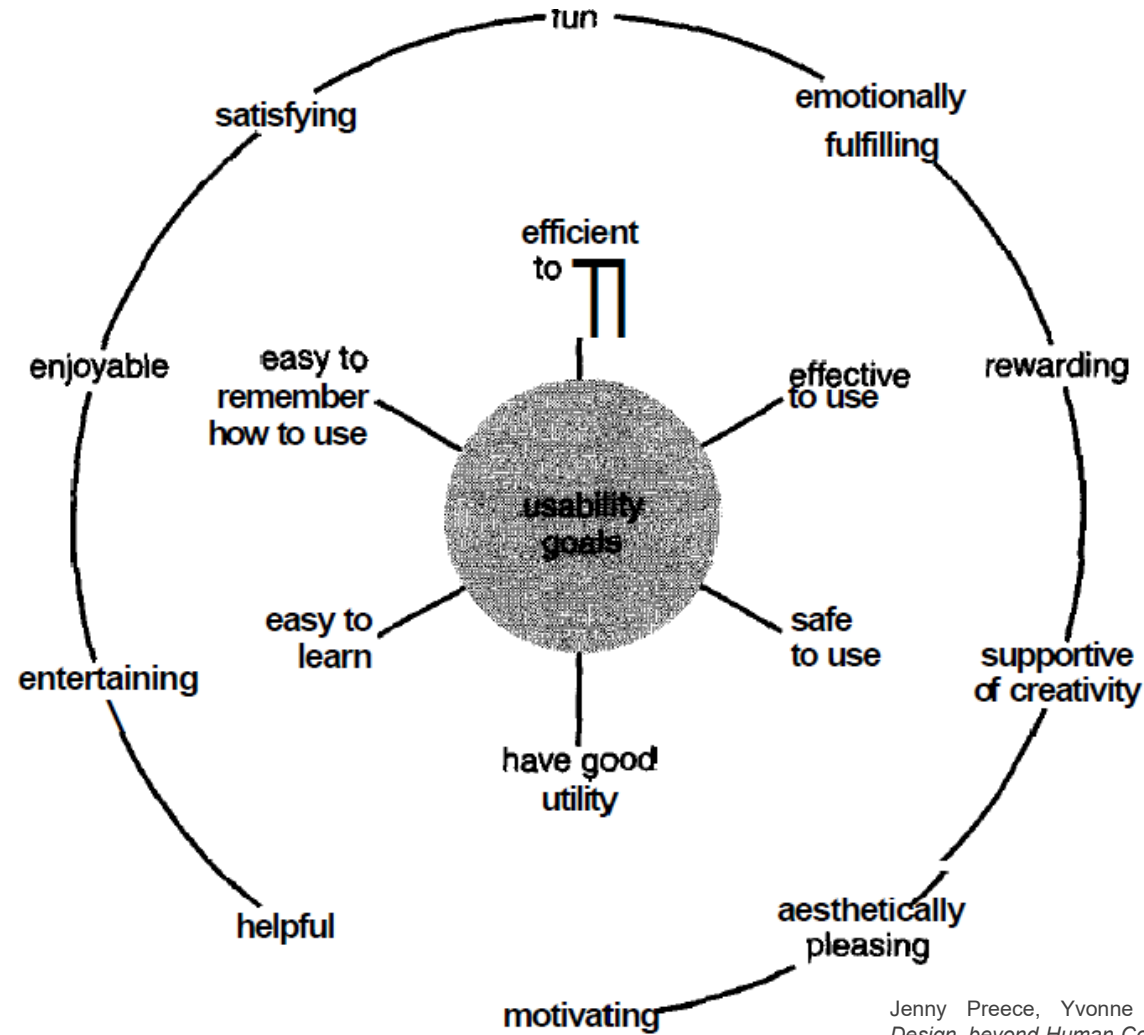
<https://s-media-cache-ak0.pinimg.com/originals/19/11/cb/1911cb9d535c9dd02ab712ea81f1ed9b.jpg>

User Interface



<http://www.sitrion.com/blog/fundamental-principles-of-self-services-ui-design>

Usability dan User Experience goals



Jenny Preece, Yvonne Rogers, Helen Sharp. 2002. *Interaction Design_beyond Human-Computer –Interaction*, J. Wiley & Sons

Tujuan

Setelah mengikuti materi ini mahasiswa dapat:

1. Mengetahui proses yang dilakukan pada metode UCD
2. Memahami pentingnya mengenal pengguna dari aplikasi yang akan dibangun sebagai dasar untuk merancang antarmuka perangkat lunak
3. Mendefinisikan Persona *User* antarmuka perangkat lunak

Overview User Centered Design (UCD)

What is User-Centered Design?

- An approach to UI development and system development.
- Focuses on understanding:
 - Users, and
 - Their goals and tasks, and
 - The environment (physical, organizational, social)
- Pay attention to these throughout development

ISO on User-centered Design

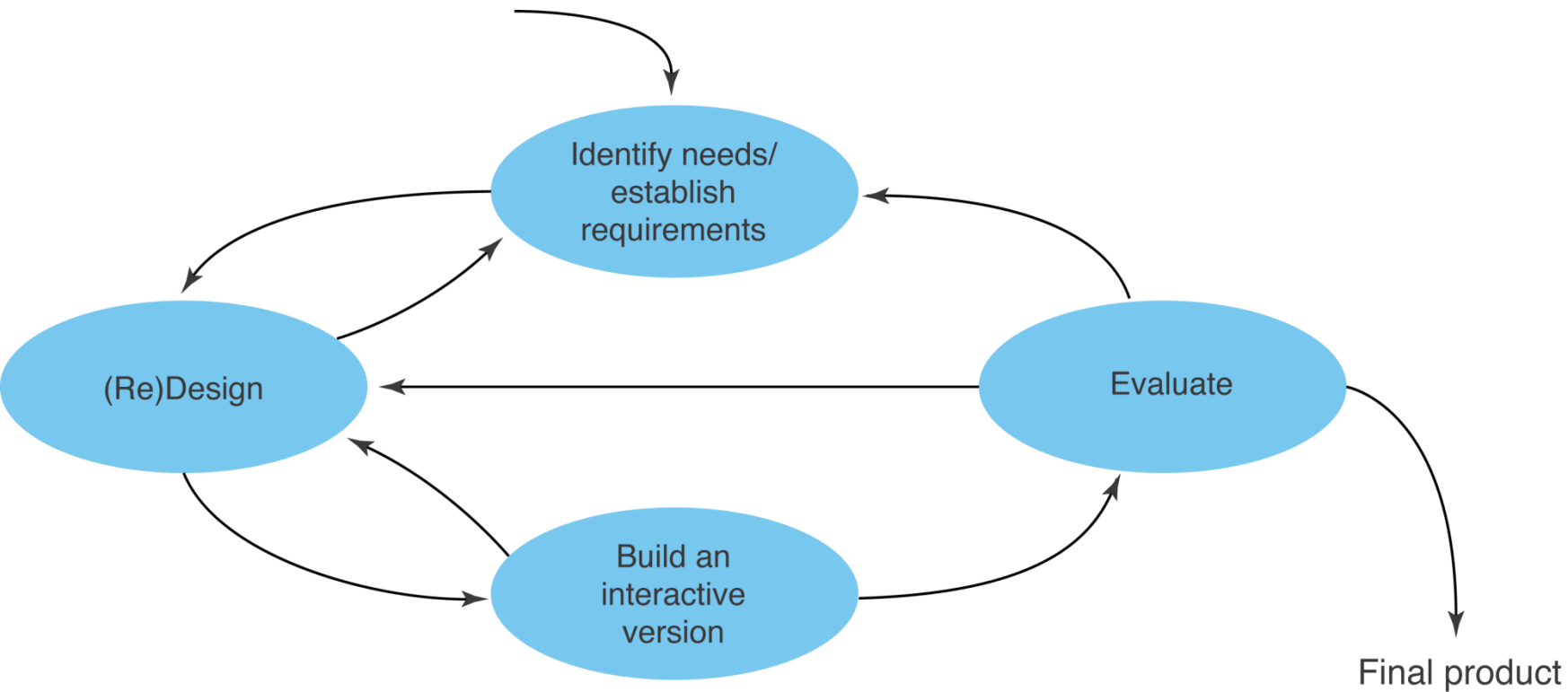
- ISO 13407 describes human-centered design processes for interactive systems
- Principles of human-centered design:
 - Active involvement of users
 - Appropriate allocation of function between user and system
 - Iteration of design solutions
 - Multidisciplinary design teams

What is a user-centered approach?

User-centered approach is based on:

- **Early focus on users and tasks:** directly studying cognitive, behavioral, anthropomorphic & attitudinal characteristics
- **Empirical measurement:** users' reactions and performance to scenarios, manuals, simulations & prototypes are observed, recorded and analysed
- **Iterative design:** when problems are found in user testing, fix them and carry out more tests

A simple interaction design model



Exemplifies a user-centered design approach

1. Identifying needs and establishing requirements

Some practical issues

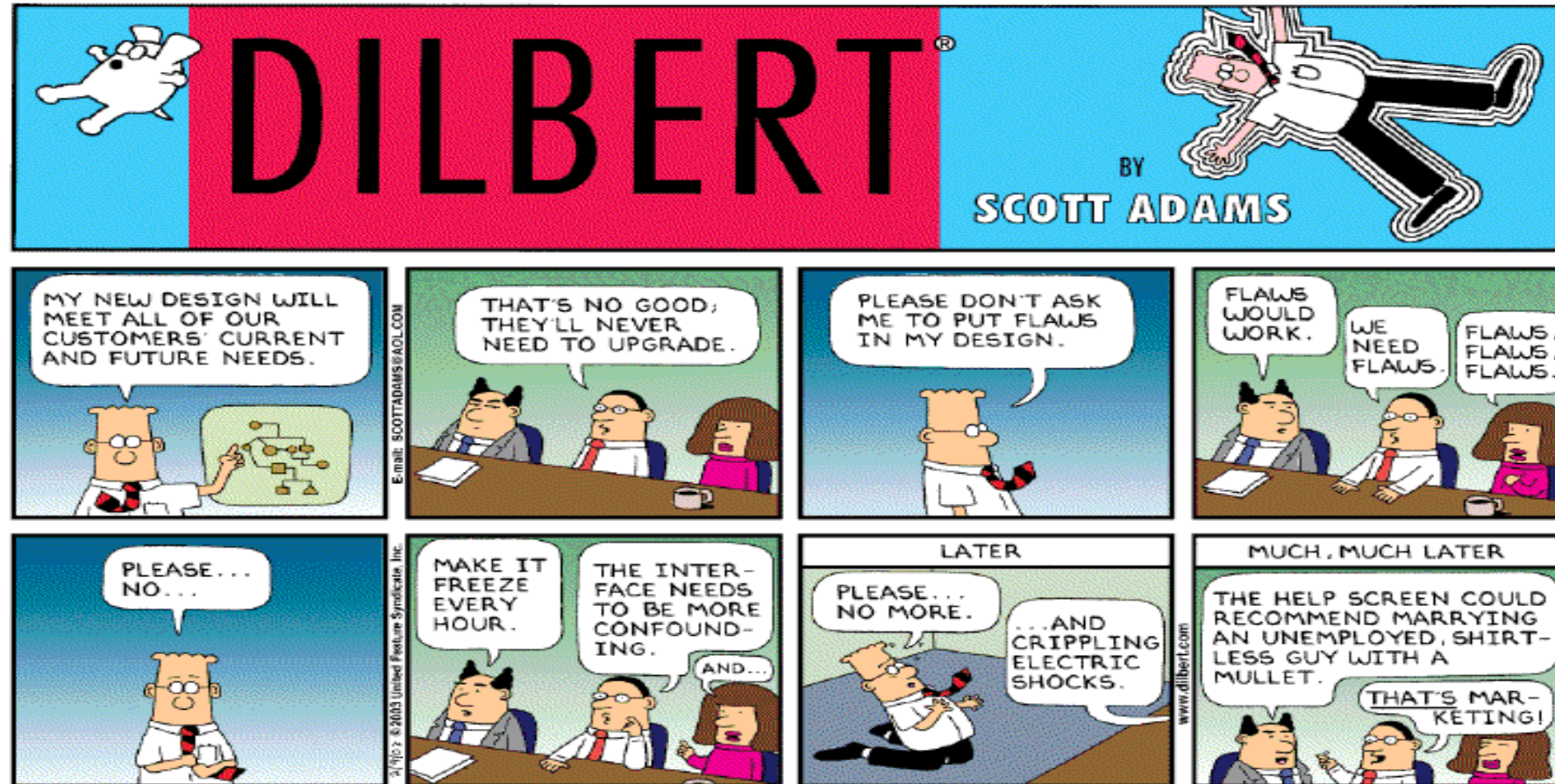
- Who are the users?
- What are 'needs'?

Target users



© Scott Adams, Inc./Dist. by UFS, Inc.

Users' needs



Brief overview of common methods to gather user data

- Interviews
- Questionnaires
- Observation
- Choosing and combining techniques

Interviews

- **Unstructured** - are not directed by a script. Rich but not replicable.
- **Structured** - are tightly scripted, a questionnaire delivered verbally. Replicable but may lack richness.
- **Semi-structured** - guided by a script but interesting issues can be explored in more depth. Can provide a good balance between richness and replicability.

Interview questions

- Two types:
 - ‘closed questions’ have a predetermined answer format, e.g., ‘yes’ or ‘no’
 - ‘open questions’ do not have a predetermined format
- Closed questions are easier to analyze
- Avoid:
 - Long questions
 - Compound sentences - split them into two
 - Jargon and language that the interviewee may not understand
 - Leading questions that make assumptions e.g., why do you like ...?
 - Unconscious biases e.g., gender stereotypes

Enriching the interview process

- Props - devices for prompting interviewee, e.g., a prototype, scenario



Contextual Inquiry

- An approach to ethnographic study. Often conducted as an apprenticeship where user is expert, designer is apprentice
- A form of interview, but
 - at users' workplace (workstation)
 - 2 to 3 hours long
- Four main principles:
 - Context: see workplace & what happens
 - Partnership: user and developer collaborate
 - Interpretation: observations interpreted by user and developer together
 - Focus: project focus to understand what to look for

Questionnaires

- Questions can be closed or open
- Closed questions are easier to analyze, and may be done by computer
- Can be administered to large populations
- Paper, email and the web used for dissemination
- Sampling can be a problem when the size of a population is unknown as is common online

Questionnaire design

- The impact of a question can be influenced by question order.
- Do you need different versions of the questionnaire for different populations?
- Provide clear instructions on how to complete the questionnaire.
- Strike a balance between using white space and keeping the questionnaire compact.
- Decide on whether phrases will all be positive, all negative or mixed.

Advantages of online questionnaires

- Responses are usually received **quickly**
- No **copying** and postage **costs**
- Data can be collected in **database** for analysis
- Time required for data analysis is **reduced**
- Errors can be corrected **easily**

Problems with online questionnaires

- Sampling is problematic if population size is unknown
- Preventing individuals from responding more than once
- Individuals have also been known to change questions in email questionnaires

Observation

- Direct observation in the field
 - Structuring frameworks
 - Degree of participation (insider or outsider)
 - Ethnography
- Direct observation in controlled environments
- Indirect observation: tracking users' activities
 - Diaries
 - Interaction logging

Structuring frameworks to guide observation

- - *The person*. Who?
 - *The place*. Where?
 - *The thing*. What?
- The Goetz and LeCompte (1984) framework:
 - *Who* is present?
 - What is their role?
 - *What* is happening?
 - *When* does the activity occur?
 - *Where* is it happening?
 - *Why* is it happening?
 - *How* is the activity organized?

Ethnography (1)

- Ethnography is a philosophy with a set of techniques that include participant observation and interviews
- Debate about differences between participant observation and ethnography
- Ethnographers immerse themselves in the culture that they study
- A researcher's degree of participation can vary along a scale from 'outside' to 'inside'
- Analyzing video and data logs can be time-consuming
- Collections of comments, incidents, and artifacts are made

Ethnography (2)

- Co-operation of people being observed is required
- Informants are useful
- Data analysis is continuous
- Interpretivist technique
- Questions get refined as understanding grows
- Reports usually contain examples



Direct observation in a controlled setting

- Think-aloud technique

Indirect observation

- Diaries
- Interaction logs

Choosing and combining techniques

- Depends on
 - The focus of the study
 - The participants involved
 - The nature of the technique
 - The resources available

2. Developing alternative designs

- **Competitive/Comparative Analysis,**
 - Try using similar services or products in order to find out:
 - Current trends in the marketplace
 - What expectations your users will have
 - What *to* do, what *not* to do
 - Interface conventions
 - “Must have” standard features
- **Heuristic Evaluation,**
 - Evaluate an existing interface (or new interface concept) based on set of usability criteria
 - Mostly used to highlight usability problems and deficiencies
 - May or may not propose usability solutions
 - Identified problem areas are addressed by subsequent design work
 - Normally done with expert evaluators, but it can be a valuable tool for anyone
 - One detailed checklist: <http://www.stcsig.org/usability/topics/articles/he-checklist.html>

2. Developing alternative designs

- **Persona,**

- Models of “archetypical” users culled from user research
- Each persona is a description of one particular “typical” user of your system
- Personas may be combined if they have the same (or sometimes overlapping) goals
- Places the focus on specific users rather than on “everyone”

- **Goals, Tasks & Scenarios,**

- **Goals:**

- Are what the user wants to do, but not how the user achieves them

- **Tasks:**

- Describe the steps necessary to achieve the goals
- Can vary with the available technology
- Are broken down into steps for task analysis, and are recombined into sequence of steps for scenario development
- Designers can reorganize, combine, or remove tasks currently performed to help users achieve their goals more efficiently

- **Scenarios:**

- Written description of a persona achieving a goal through a set of tasks in a specific context
- Should start technology-neutral and become more specific as the design progresses

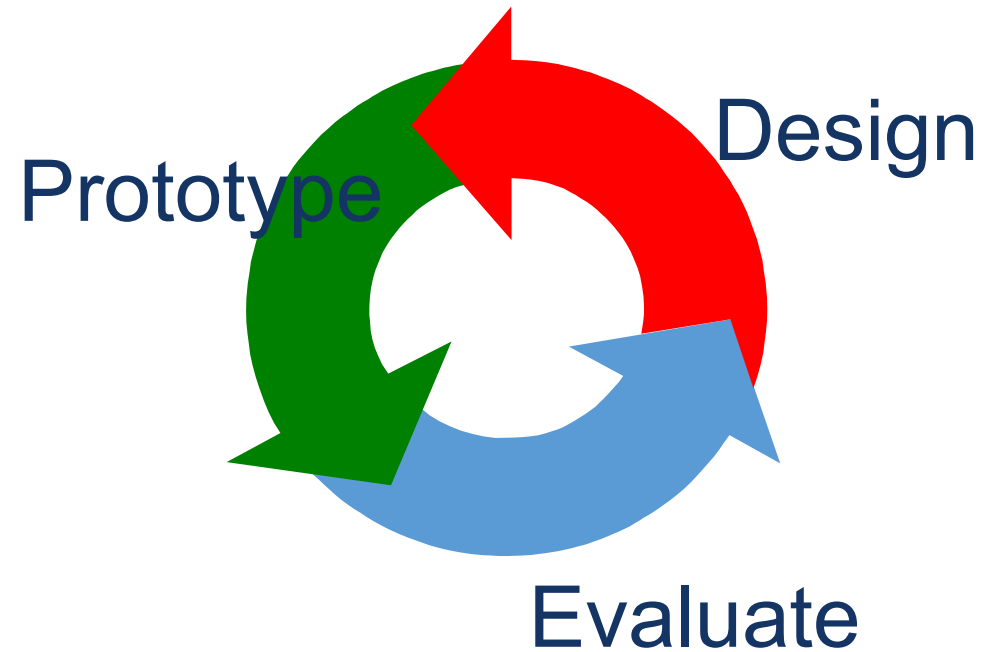
What is Persona

<http://webdesign.tutsplus.com/articles/defining-and-applying-personas-to-ux-design--webdesign-7561>

- In terms of web design, a persona is a fictional example of a person within your user base. It typically includes details such as
 - age
 - sex
 - occupation
 - hobbies
 - likes/dislikes
 - other details germane to the product
- Many times it also includes the personalities of these fictional users. The goal of erecting these details is to establish the mindset, desires, and necessary tasks of your users. Ideally, the persona creation process is informed by research and/or an intimate knowledge of an established user base. How you design them and what information you include is up to you and what you need

3. Building interactive versions of the designs

- Start rough
- Explore!
- Use personas to keep the users in view
- Use scenarios to inform the design
- Get frequent feedback
- Note user conventions
- Make design artifacts public
- May be expressed in a prototype for usability testing



4. Evaluating designs

- Let users validate or invalidate the design
- Ask the user to complete selected typical tasks (from scenarios) and think aloud while they do it
- Test early in the process
- Can test with 3-5 users (or less!)
- “Formal” testing
- Measures “success”
 - Set success criteria prior to testing (best done at the project outset)
 - Compare to baseline if you have one
 - Have usability problems revealed in the heuristic evaluation been addressed?

4. Evaluating designs

- Define what is to be tested
- Select users based on personas
- Administer the tests
- Analyze the data
- Document the findings in a brief
- Share the findings with the development team
- Determine what design changes will be made based on test results

THE USER INTERFACE (UI) DESIGN PROCESS

Step 1 : Know Your Users & Clients

Step 1 : Know your users and clients

- Understand how people interact with computers.
- Understand the human characteristics important in design.
- Identify the user's level of knowledge and experience.
- Identify the characteristics of the user's needs, tasks, and jobs.
- Identify the user's psychological characteristics.
- Identify the user's physical characteristics.
- Employ recommended methods for gaining understanding of users.

Output Step 1

- Adapun output dari step 1 ini adalah menghasilkan **Persona *User***
- Persona *User* dibuat untuk :
 - Help identify specific users and needs
 - Identify primary, secondary users
 - Helps choose the right individuals to design for

Understand how people interact with computers

- Siklus aksi manusia :
 - Tujuan dibentuk
 - Sebuah rencana eksekusi disusun dan diimplementasikan
 - Hasil dari sebuah tindakan kemudian dievaluasi
- Mengapa orang memiliki masalah dengan komputer
 - Terlalu banyak fleksibilitas
 - Menggunakan jargon
 - Desain yang membingungkan/tidak jelas
 - Perbedaan yang terlalu halus
 - Disparitas dalam strategi pemecahan masalah

Understand the human characteristics important in design

- Respon psikologis yang khas untuk desain buruk adalah:
KEBINGUNGAN, ANNOYANCE, FRUSTRASI, PANIC / STRES, KEBOSANAN.
- Kemudian akan diikuti dengan reaksi fisik :
 - Pengabaian sistem.
 - Penggunaan sebagian dari sistem.
 - Penggunaan langsung dari sistem.
 - Modifikasi tugas.
 - Aktivitas kompensasi.
 - Penyalahgunaan sistem.
 - Pemrograman langsung

Understand the human characteristics important in design

- Perception
- Memory : Short term memory (STM) dan Long term memory (LTM)
- Sensory storage
- Visual Acuity
- Information Process
- Mental Models
- Movement Control
- Learning
- Skill
- Performance Load
- Individual Differences

Understand the human characteristics important in design

- Faktor penentu utama untuk merancang sistem interaktif
- Pemahaman bagaimana orang berpikir, belajar dan berkomunikasi
- Bisa dianalogikan dengan model pemrosesan dalam komputer

3 Phase Human Information Processing System

1. **Perceptual System (Read – scan)**
2. **Cognitive System (Think)**
3. **Motor System (Response)**

The Human Information Processing System

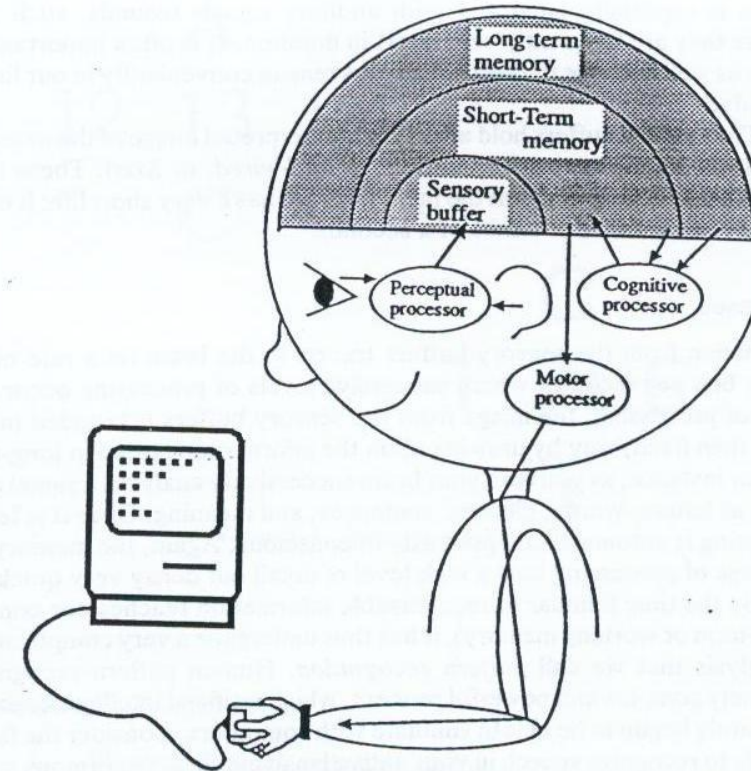


Figure 2-3 The human information processing system. From Card, "The Model Human Processor: A Model for Making Engineering Calculations of Human Performance," *Proceedings of the Human Factors Society 25th Annual Meeting*, 1981, 301-5. © 1981, by the Human Factors Society Inc. Reprinted with permission. All rights reserved.

Perceptual System

Memory Structure :

- Sensory organ → sensory buffer to hold a virtual, uninterpreted image of external signal
- Can hold a high level of detail, but a very short in duration
- Information will decay or travel to brain

Perceptual Process

- Image from sensory buffer is recorded in a learned way by drawing upon the information store in long-term memory
- Human Pattern Recognition is not a simple matter of matching a pattern it makes heavy use of context and knowledge
- Human Pattern Recognition is a active processing, able to tolerate an ambiguity and incompleteness information.

Cognitive System

Short-term Memory (STM) :

- Not only a storage buffer but also a (very slow) processor
- Limited in Capacity and Duration

Long Term Memory (LTM) :

- Unlimited in Capacity and permanent duration
- Request effort & concentration to put new information in LTM

Motor System

- It is believe that thought is translated into action through a series of discrete micromovement
- Motor movement relevant to UI design include :
head, eye, arm, hand and finger movement

Human Considerations in the Design of Business Systems

- The User's Knowledge and Experience
- The User's Tasks and Needs
- The User's Psychological Characteristics
- The User's Physical Characteristics

Identify the user's level of knowledge and experience

KNOWLEDGE/EXPERIENCE	
Computer Literacy	Highly technical or experienced, moderate computer experience, or none.
System Experience	High, moderate, or low knowledge of a particular system and its methods of interaction.
Application Experience	High, moderate, or low knowledge of similar systems.
Task Experience	Level of knowledge of job and job tasks.
Other Systems Use	Frequent or infrequent use of other systems in doing job.
Education	High school, college, or advanced degree.
Reading Level	Less than 5th grade, 5th–12th, more than 12th grade.
Typing Skill	Expert (135 WPM), skilled (90 WPM), good (55 WPM), average (40 WPM), or "hunt and peck" (10 WPM).
Native Language or Culture	English, another, or several.

Identify the characteristics of the user's needs, tasks, and jobs

JOB / TASK / NEED	
Type of System Use	Mandatory or discretionary use of the system.
Frequency of Use	Continual, frequent, occasional, or once-in-a-lifetime use of system.
Task or Need Importance	High, moderate, or low importance of the task being performed.
Task Structure	Repetitiveness or predictability of tasks being automated, high, moderate, or low.
Social Interactions	Verbal communication with another person required/not required.
Primary Training	Extensive or formal training, self-training through manuals, or no training.
Turnover Rate	High, moderate, or low turnover rate for jobholders.
Job Category	Executive, manager, professional, secretary, clerk.
Lifestyle	For Web e-commerce systems, includes hobbies, recreational pursuits, and economic status.

Identify the user's psychological characteristics

PSYCHOLOGICAL CHARACTERISTICS	
Attitude	Positive, neutral, or negative feeling toward job or system.
Motivation	Low, moderate, or high due to interest or fear.
Patience	Patience or impatience expected in accomplishing goal.
Expectations	Kinds and reasonableness.
Stress Level	High, some, or no stress generally resulting from task performance.
Cognitive Style	Verbal or spatial, analytic or intuitive, concrete or abstract.

Identify the user's physical characteristics

PHYSICAL CHARACTERISTICS	
Age	Young, middle aged, or elderly.
Gender	Male or female.
Handedness	Left, right, or ambidextrous.
Disabilities	Blind, defective vision, deafness, motor handicap.

An Example Persona : Marjorie Bannet

- **Biography**

- 78 years old
- Just moved to Penrith from Windermere
- Has a son in Hastings, and a daughter in Newcastle
- Doesn't know anyone else in Penrith yet
- Hasn't been driving for a few years now
- Sometimes feels lonely
- Has a help come in once a week
- Would like to be able to read more



An Example: Marjorie Bannet

- **Health**

- Has trouble sleeping from time to time. Will wake up in the early hours and often not get to sleep again for 2-3 hours
- A little arthritis in her hands
- Early cataracts, so less acute vision
- Can move about, perhaps not quite as quickly as she could 10 years ago
- Sometimes has a rest in the afternoon



An Example: Marjorie Bannet

- **Technology**

- Has never used a computer before, and is a little nervous about them
- Has a mobile phone, and instructions on how to use it from her son
- Uses the microwave to prepare many of her meals
- Uses a video recorder, but can't be bothered setting it to record things



User Persona Example

TOBI DAY



PERSONA TEMPLATE

AGE 26

OCCUPATION Record Store Manager

STATUS Single

LOCATION New York, NY

TIER Enthusiast

ARCHETYPE The Maestro

Ambitious **Admired** **Focused**



"If I had a way to share projects and collaborate in real time, that would make my workload so much easier to manage."

MOTIVATIONS

Incentive	
Fear	
Achievement	
Growth	
Power	
Social	

GOALS

- To grow a strong industry reputation
- To build an audio-pro portfolio
- To keep track of everything

FRUSTRATIONS

- Slow download times
- Data crashes
- Poor communication

BIO

Tobi has a day job at a record store, but on the side she does all kinds of production work for up-and-coming artists. She never hesitates to learn something new and she often acts as tech support for her friends and clients. She is usually working on a dozen projects at a time and is trying to establish herself in the industry, so she hates data crashes or anything that makes her look bad. Because she works alone and in her home, collaboration is everything.

PERSONALITY

Extrovert	Introvert
Sensing	Intuition
Thinking	Feeling
Judging	Perceiving

TECHNOLOGY

IT and Internet	
Software	
Mobile Apps	
Social Networks	



User Persona Example

Clark Andrews

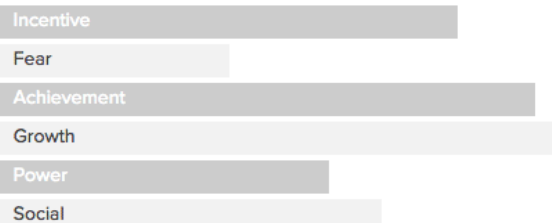
AGE 26
OCCUPATION Software Developer
STATUS Single
LOCATION San Jose, CA
TIER Experiment Hacker
ARCHETYPE The Computer Nerd

Friendly Clever Go-Getter



"I feel like there's a smarter way for me to transition into a healthier lifestyle."

Motivations



Goals

- To cut down on unhealthy eating and drinking habits
- To measure multiple aspects of life more scientifically
- To set goals and see and make positive impacts on his life

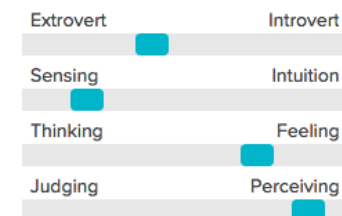
Frustrations

- Unfamiliar with wearable technology
- Saturated tracking market
- Manual tracking is too time consuming

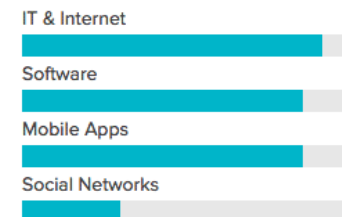
Bio

Aaron is a systems software developer, a "data junkie" and for the past couple years, has been very interested in tracking aspects of his health and performance. Aaron wants to track his mood, happiness, sleep quality and how his eating and exercise habits affects his well being. Although he only drinks occasionally with friends on the weekend, he would like to cut down on alcohol intake.

Personality



Technology



Brands



User Persona Example

Background:

- *Staff Accountant at Founder Accounting
- *Completed his undergraduate degree at Penn State and his masters at Cornell University
- *Has a serious girlfriend and two dogs (a Labrador retriever and a pug mix)

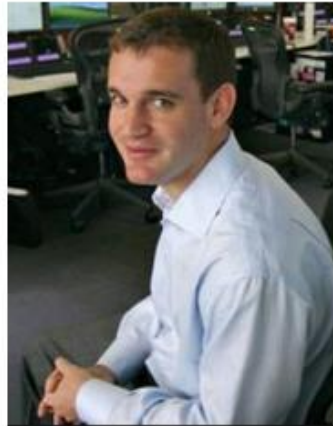
Demographics:

- *Male
- *Age 34
- *Annual HH income: \$125,000
- *Lives in a townhouse-style condo in an urban area

Goals:

- *Become a senior accountant within 3-5 years
- *Achieve a salary of \$80,000 so that he can purchase a single family home
- *Network aggressively in order to build professional contacts

Tommy Technology



Hobbies & Interests:

- *Running 5K races with his girlfriend
- *Watching Game of Thrones
- *Going out to brunch with other young couples
- *Taking one nice vacation a year to established tourist destinations

Challenges:

- *Wants to have a more modern website, but isn't the final decision-maker
- *Struggles with being seen as the "young guy" in the office and being taken less seriously as a result

Common Objections:

- *I love the idea of a new website, by my boss will never go for it! He doesn't see the value in new technology.
- *I'd love to get started on a new website, but I don't think I can get buy-in from my boss. He never takes my ideas seriously.

Biggest Fears:

- *Getting stuck in a job and not advancing up the corporate ladder as quickly as he'd like
- *Economic recessions that mean he'll never be able to retire
- *Life passing him by too quickly

User Persona Example



Techie

Mike

25-34 single male
living with friends

Mike works as a graphic designer in a small agency and one day wants to run his own agency.

He's got an iPhone and a Vodafone 360 H1 by Samsung phone, one personal and one for work. He follows friends and key people in the design industry via Twitter, blogs, and RSS feeds. He uses his iPhone for work emails and his H1 for Facebook.

He uses Twitter to post updates about what he's up to with his project work as well as using it as a tool to find out what people are up to and to invite them to events. He uses Facebook to share personal photos and video and keeps a Tumblr blog to post interesting things he discovers and share them with his friends and followers.



Socialite

Zoë

18-33 single female
living with friends

Zoë is studying a Masters in International Development unsure of what the future lies ahead of her.

She is constantly using the Facebook app on her Vodafone 360 M1 by Samsung phone as well as on her PC to upload and tag photos and videos from places she's been to with her friends, as well as to find out and comment on who's been where at which club nights and parties.

She regularly texts and messages her friends to find out if they've heard about a new pop-up shop she heard about via a flyer, or one-off warehouse party started by friends of friends.



Cost-conscious

Geoff

35-49 married male
with young kids

Geoff works as an senior architect in a large practice, and has a wife and a young girl and 6-month baby boy. He thinks the time is right to start looking for a bigger home for his family.

Geoff uses his Vodafone 360 H1 to take photos and videos of prospective sites he visits. He purchased the H1 because of its ability to check email, surf the web, use apps, and take photos and video.

He loves the built-in camera and also uses this phone on holiday to take snaps of the family as it fits in his pocket and doesn't want to carry a large SLR around with him. He likes to upload his photos and video to Flickr and share them with his family and friends. He also creates photo books from his holidays snaps to give as gifts to his parents.

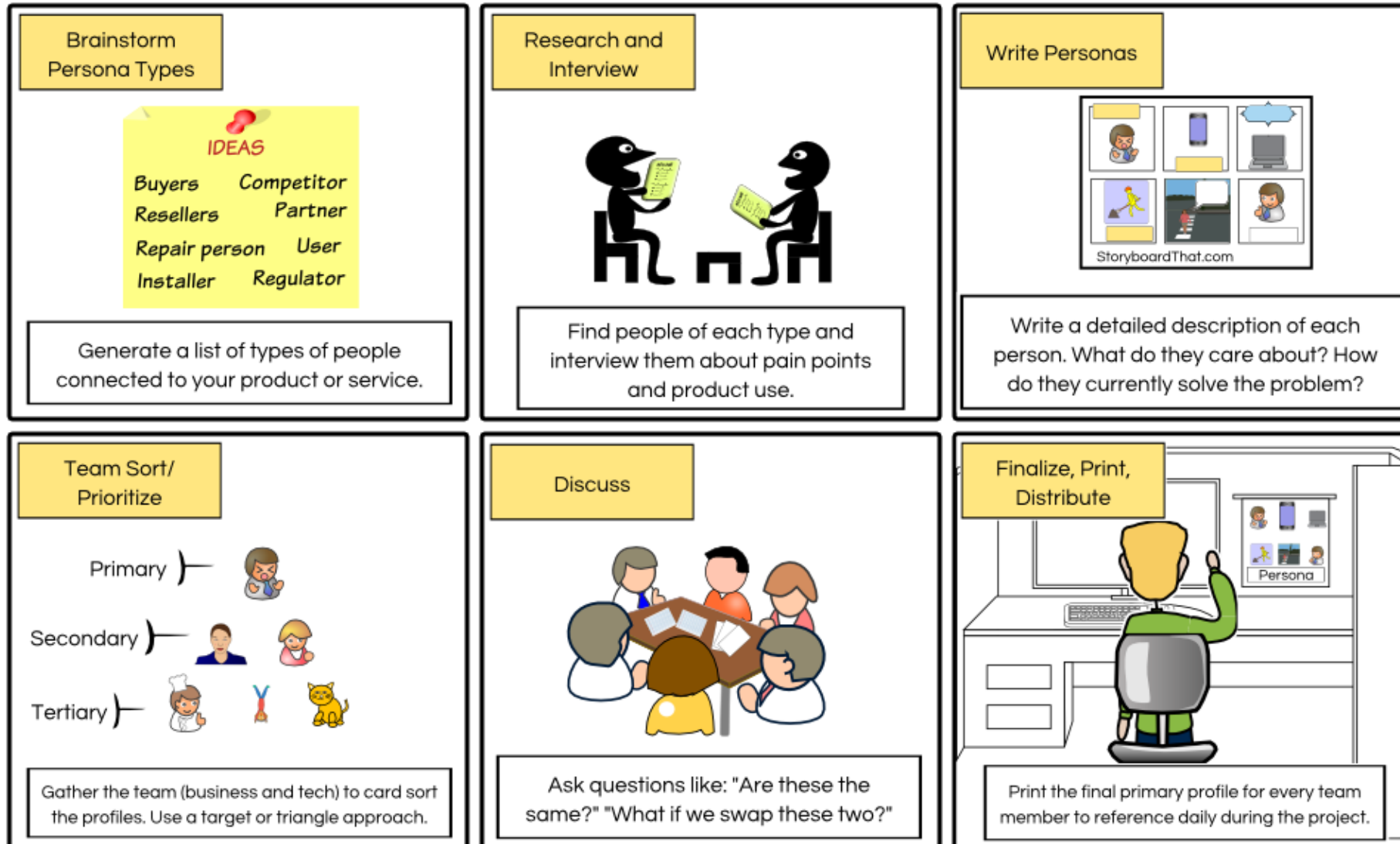
User Persona Example

			
Rodger Rejector	Tricia Traditionalist	Simon Start-Up	Sophisticated Simon Stuck
▲ Seeking to grow business	▲ Seeking to grow business	▲ Start-up	▲ Seeking to grow business
● Does not have a website	● Has a very old, basic website	● Don't have a website	● Has a website
	☹ Not happy with it but don't know how to change		☹ Not happy with it
	Can change today		Stuck for 6 months in contract
Not actively seeking to buy but wanting to understand more about websites	Actively searching for new site but doubtful and worried	Actively searching for a website	Not actively searching for site now but starting to research possibilities
Searching for advice on websites			Searching for how to improve current site especially searching around how to use SEO, Social and analytics to improve site performance
? No/ basic digital knowledge	? No/ basic digital knowledge	? Intermediate digital knowledge	? Advanced digital knowledge
No relationship with Yell	Relationship with Yell (YP)	No relationship with Yell	No relationship with Yell
Doesn't know Yell does digital products/ websites	Doesn't know Yell does digital products/ websites	Doesn't know Yell does digital products/ websites	Doesn't know Yell does digital products/ websites

Tahapan untuk dapat mendefinisikan Persona (User)

- Kunjungi lokasi pengguna, terutama jika tempat kerja pengguna yang tidak familiar untuk kita. Hal ini untuk menuntun kita mengetahui lingkungan kerja pengguna.
- Berbicaralah dengan para pengguna untuk mengetahui apa masalah mereka, kesulitan, harapan serta apa yang membuat mereka nyaman saat ini.
- Membangun komunikasi secara langsung dengan pengguna. Hindari perantara.
- Melakukan observasi bagaimana cara mereka melakukan setiap task serta melihat apa kesulitan dan masalah mereka.
- Merekam cara kerja pengguna dalam melakukan task dan mempelajari masalah dan kesulitannya.
- Mempelajari bagaimana sebuah alur kerja dalam lingkungan dimana sistem akan diinstal.
- Apakah pengguna berpikir keras ketika mereka melakukan sesuatu untuk mengungkapkan rincian yang tidak mungkin dan hal lain yang dapat dilakukan.
- Cobalah sendiri melakukan kegiatan pengguna.
- Melakukan survey dan kuesioner untuk mendapatkan sampel pengguna.
- Buat sebuah tools/alat untuk dapat menggali perilaku user dalam mencapai tujuannya, kemudian ukur sampai dengan kebutuhannya tercapai.

Steps to Create Personas



How to Create User Persona

- How To Create UX Personas →
<https://www.youtube.com/watch?v=B23iWg0koi8>
- Personas and User Flows UXUI Design Product Design Udacity →
<https://www.youtube.com/watch?v=wClXAzTFUIY>
- Creating personas for effective user stories →
<https://www.youtube.com/watch?v=tCAeHfvsjoM>

Template

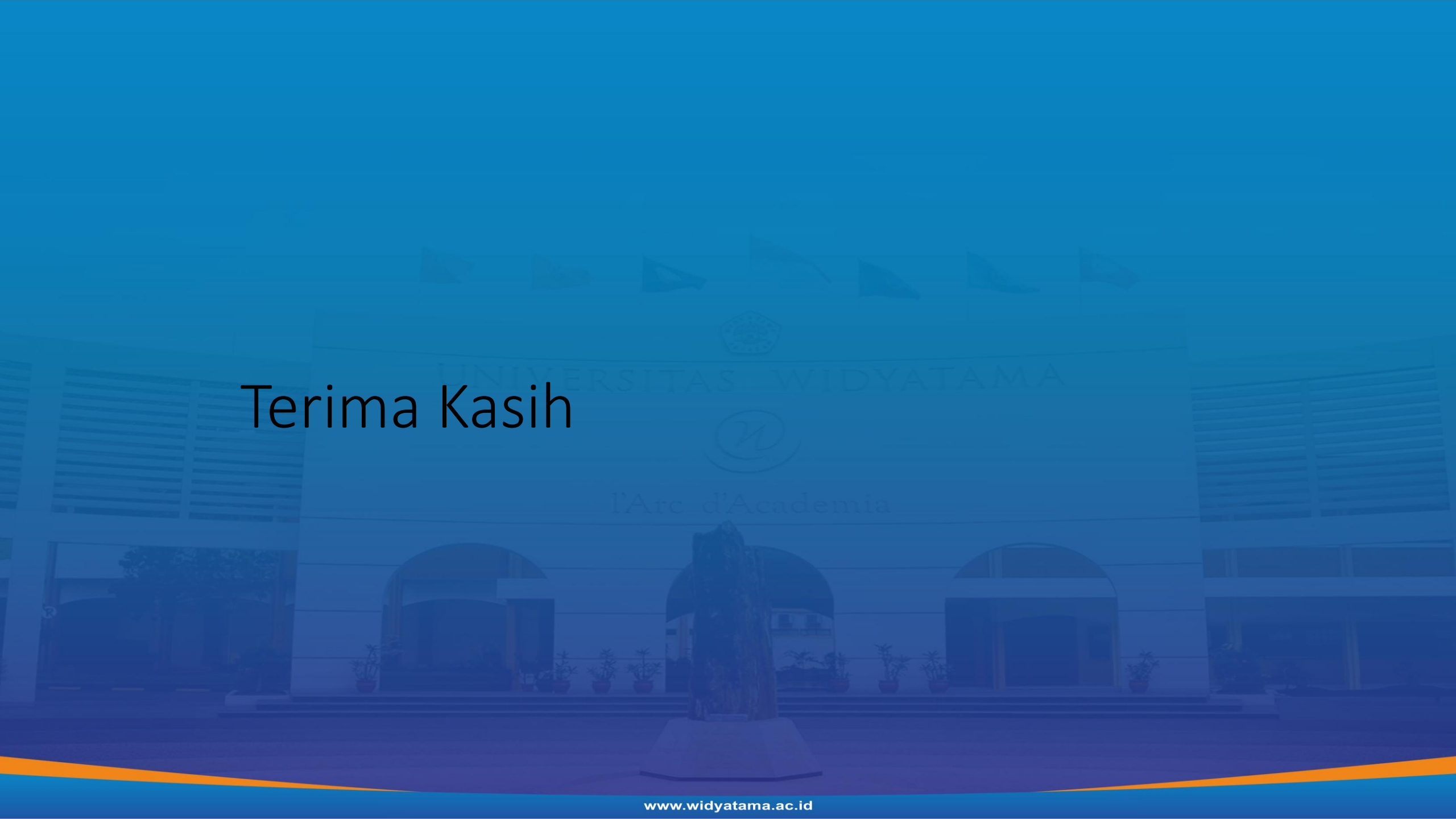
- <http://www.ux-lady.com/diy-user-personas/>
- <https://www.pinterest.com/kooj/personas/>

Tools

- <http://xtensio.com/user-persona/>
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Bahan bacaan

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